Md Abid Al Morshed | Computational Mechanics | Composite Fracture

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CORE COMPETENCIES

Mechanical Engineering PhD candidate with experience in **Abaqus FEA, structural simulation, and failure analysis**. Skilled in custom FEA implementation (**UEL/UMAT**), composite material modeling, and simulation-to-experiment comparison. Seeking an internship in **CAE, structural mechanics, or mechanical engineering R&D**.

EXPERIENCE

Graduate Research Assistant | Fall 2024–Present **Michigan Technological University**

- Developed a 3D phase-field model for composite fracture under static and dynamic loading.
- Implemented the formulation through custom Fortran user elements in Abaqus.
- Publication Status: 2 journal papers under review; 1 conference paper accepted (ASC 2026).

Graduate Teaching Assistant | Spring 2025 **Michigan Technological University**

- Led practical sessions in control engineering and HVAC analysis.
- Taught MATLAB/Simulink and Python for engineering modeling and data analysis.

Adjunct Faculty | Spring 2024 | Dept. of Industrial and Production Engineering **Bangladesh University of Textiles**

- Courses taught: Automation & Control; Mechanical Engineering & Manufacturing.

Engineering Intern | Fall 2022 **Energypac Engineering Ltd.**

- Completed industrial training in transformer manufacturing, quality control, and substation operations.

ACTIVE PROJECTS

1. Coupling intralaminar fracture and delamination within a mixed-mode phase-field framework.
2. Investigating layer orientation and stacking effects on fracture resistance in 3D-printed composites (Collaboration)

EDUCATION

Michigan Technological University | GPA: 3.84 Expected graduation: Fall 2027

Ph.D. in Mechanical and Aerospace Engineering | Computational modeling of composite fracture

Bangladesh University of Engineering and Technology | GPA: 3.69 Graduated: May 2023

B.Sc. in Mechanical Engineering | Computational fluid mechanics and heat transfer

ENGINEERING PROFICIENCIES

Engineering analysis: Nonlinear FEA, structural failure analysis, fracture mechanics, composite failure.

Software & Programming: Abaqus (custom user elements: UEL/UMAT), Fortran, Python, MATLAB-Simulink, CAD.

HONORS AND PROFESSIONAL INVOLVEMENTS

Awards: College of Engineering Candidacy Incentive Award, Michigan Tech (2026); Dean's List and University Merit Scholarship, BUET (2017, 2018).

Conference Presentations: American Society for Composites (2026); Society of Engineering Science (2026).

Research Communication: Three-Minute Thesis, MTU (2025, 2026), ICC computing showcase (2026).

Community Involvement: Research Rotation Mentor, MTU (2026); **Volunteer**: Campus events and trail cleanup (2025–2026).

SELECTED PUBLICATONS

Journal Paper

- **MAA Morshed**, NA Mini, MA Hakim, and MN Hasan (2025). "Mixed convective heat transfer characteristic in vented cavity under dynamic flow modulation by CFD and neural network model approaches". *Heat Transfer, 54*(5), 3072–3087. DOI:10.1002/htj.23349

Book Chapter

- **MAA Morshed** and NA Mini. "Lessons Learned from CFRP Failures: Case Studies and Insights." In Handbook of Carbon Fiber Reinforced Polymer Composites, edited by Md Zillur Rahman and Kalim Deshmukh. *Springer Nature*, (accepted), 2026.

Conference Proceeding

- **MAA Morshed**, MA Mahmud, M Atiqur, and MZ Haq. "Thermal Analysis of an Advanced Adiabatic Compressed Air Energy Storage System with Phase Changing Materials for Thermal Energy Storage." *Proceedings of the Energy Conference 2023: National and Global Issues (ENCON23)*, 2024. DOI:10.2139/ssrn.4993585